

BASIC SIGHT-SINGING AND EAR TRAINING FOR CHURCH SINGERS

UNIT 4.2

PITCH

BASIC DEFINITIONS:

INTERVALS IN THE MAJOR SCALE.

CLEFS AND KEY SIGNATURES.

THE KEY OF C.



GETTING TO KNOW THE INTERVALS WITHIN THE COMPLETE MAJOR SCALE

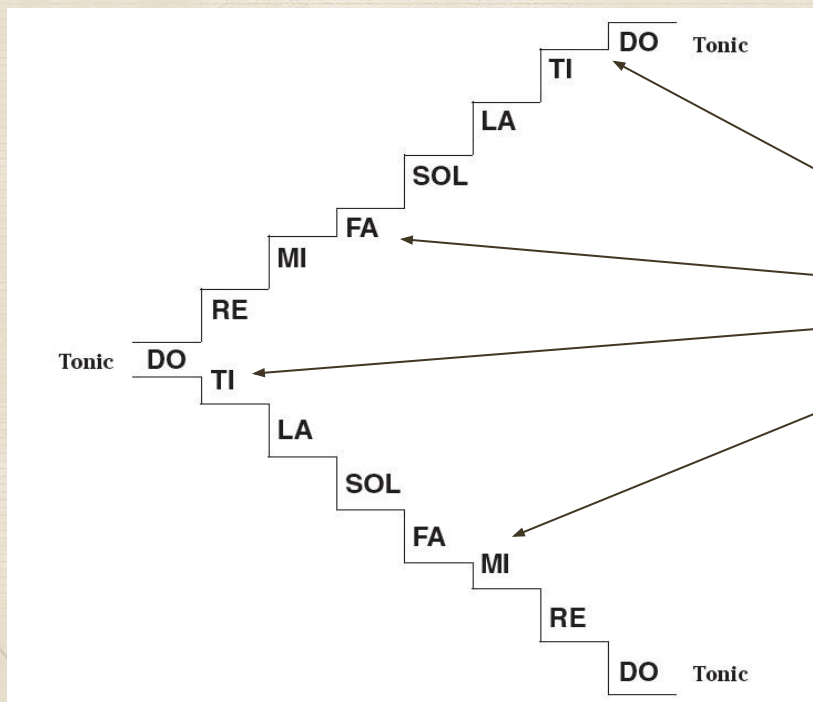
In Unit 3.2 (Slides 7, 8, & 9) we saw how combining the DO-FA and SOL-DO Tetrachords, one after the other, forms the complete **major scale**. By definition, the major scale has the following arrangement of **whole steps** and **half steps** (W=whole step, H=half step):

	W		W		H		W		W		W		H	
DO	-	RE	-	MI	-	FA	-	SOL	-	LA	-	TI	-	DO

Changing this arrangement of intervals **in any way** would make the scale no longer major; it would become some other type of scale (called a “**mode**”), as will be discussed in a later unit. Various modes are encountered in Byzantine and other types of liturgical chant, such as Georgian, Serbian, Gregorian, etc.

“LA SCALA”

The major scale—ascending and descending—can be depicted using the following diagram.
(**Scala** in Italian, from which we get the word “scale,” means “ladder” or “staircase.”)



Notice that the size of the steps from MI to FA and from TI to DO are **smaller** than all the rest. This is because they are **half steps**.

A TABLE OF INTERVALS IN THE MAJOR SCALE

The various intervals found within the major scale are summarized below. This chart can be regarded as the sight-singer's "multiplication tables," to be practiced daily and memorized.

1. Seconds

DO	RE	DO	Major Second (whole step)
RE	MI	RE	Major Second (whole step)
MI	FA	MI	Minor Second (half step)
FA	SOL	FA	Major Second (whole step)
SOL	LA	SOL	Major Second (whole step)
LA	TI	LA	Major Second (whole step)
TI	DO	TI	Minor Second (half step)
DO			

3. Fourths

DO	FA	DO	Perfect fourth (whole, whole, half)
RE	SOL	RE	Perfect fourth (whole, half, whole)
MI	LA	MI	Perfect fourth (half, whole, whole)
FA	TI	FA	Augmented fourth (tritone) (whole, whole, whole)
SOL	DO	SOL	Perfect fourth (whole, whole, half)
LA	RE	LA	Perfect fourth (whole, half, whole)
TI	MI	TI	Perfect fourth (half, whole, whole)
DO			

2. Thirds

DO	MI	DO	Major third (whole step, whole step)
RE	FA	RE	Minor third (whole step, half step)
MI	SOL	MI	Minor third (half step, whole step)
FA	LA	FA	Major third (whole step, whole step)
SOL	TI	SOL	Major third (whole step, whole step)
LA	DO	LA	Minor third (whole step, half step)
TI	RE	TI	Minor third (half step, whole step)
DO			

4. Fifths

DO	SOL	DO	Perfect fifth (whole, whole, half, whole)
RE	LA	RE	Perfect fifth (whole, half, whole, whole)
MI	TI	MI	Perfect fifth (half, whole, whole, whole)
FA	DO	FA	Perfect fifth (whole, whole, whole, half)
SOL	RE	SOL	Perfect fifth (whole, whole, half, whole)
LA	MI	LA	Perfect fifth (whole, half, whole, whole)
TI	FA	TI	Diminished fifth (tritone) (half, whole, whole, half)
DO			

MORE INTERVALS IN THE MAJOR SCALE

Within the major scale we also can form larger intervals such as **Minor and Major Sixths**, as well as **Minor and Major Sevenths**, and the interval of the **Augmented Fourth** (or **Diminished Fifth**), also known as the **Tritone** (FA-TI, TI-FA—formed by three successive whole steps). The latter was so scrupulously avoided in chant that medieval music theorists called it “*Diabolus in musica*” (the Devil in music). These intervals are summarized below:

MI-DO↑, TI-SOL↑, LA-FA↑

Minor Sixths (formed by stepping 5 notes from the starting note—a total of 2 half steps and 3 whole steps)

DO-LA↑, RE-TI↑, FA-RE,
& SOL-MI↑

Major Sixths (formed by stepping 5 notes from the starting note—a total of 4 whole steps and a half step)

RE-DO↑, MI-RE↑, SOL-FA↑,
LA-SOL↑, & TI-LA↑

Minor Sevenths (formed by stepping 6 notes from the starting note—a total of 4 whole steps and 2 half steps)

DO-TI↑, FA-MI↑

Major Sevenths (formed by stepping 6 notes from the starting note—a total of 5 whole steps and a half step)

We have introduced all the various intervals in the major scale to make a complete presentation. Practically speaking, in singing Orthodox church melodies we encounter:

most often

- ◆ half steps
- ◆ whole steps
- ◆ major and minor thirds

occasionally

- ◆ perfect fourths
- ◆ perfect fifths
- ◆ octaves

even more rarely

- ◆ major and minor sixths

practically never

- ◆ tritones
- ◆ minor and major sevenths

INTRODUCING COMMON WESTERN STAFF NOTATION

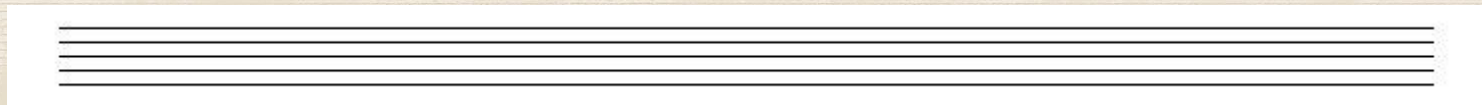
Common Western **staff notation** evolved over the past 1000 years as a way of representing visually, in **two-dimensional space**, the movement of a melody up or down pitch-wise (vertical dimension or “Y-axis”) as it moved through time (horizontal dimension or “X-axis”).

Centuries ago, Orthodox church singers began using, and in some cases, continue to use some type of staffless “sign” notation, such as Byzantine neumes (shown in Rhythm exercise 3.1, Slide 8) or Russian “znamenny” notation (shown in Rhythm exercises 1.1, Slide 8) from the word *znamia*, meaning “sign”. The “signs” (also known as **neumes**) told the singers, who already had memorized dozens of notational signs by heart, what melodic movement to execute with their voice.

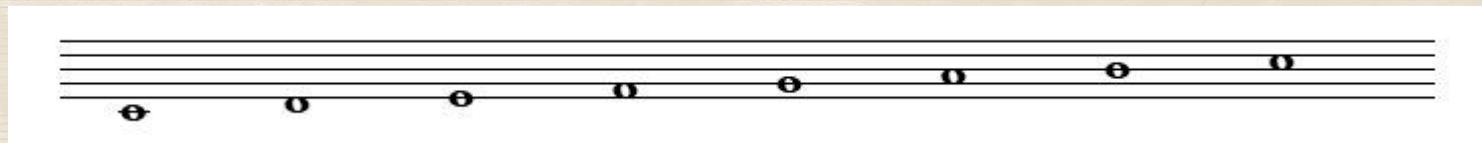
But in the end, staff notation proved to be more convenient because of its **graphic representational nature** in two dimensions: up and down, and left to right. Among the Eastern Slavs (Russians and Ukrainians), staff notation gradually supplanted the staffless znamenny notation in the 17th century. The latter is still used to this day by Russian “Old Ritualists” or “Old Believers.”

THE MODERN-DAY FIVE-LINE STAFF

Rather than use the staircase model, **round notes** (points), representing pitches and their relative durations (see Unit 1), were arranged on a series of lines called a **staff**. Gregorian Chant staff notation used a four-line staff; Russian “Kievan” notation (see Unit 4.1, Slide 9), which dates back to the late sixteenth century, used a five-line staff, as does modern-day notation.



But simply arranging a series of notes (in this case, whole notes) on a five-line staff does not yet tell us what pitches they represent or what is the relationship between them.

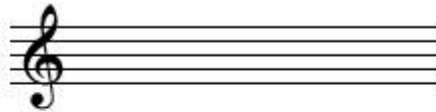


CLEFS AND KEY SIGNATURES

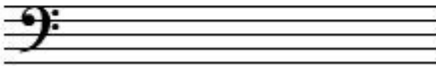
Two things are needed to “unlock” the meaning of the pitches shown by the notes on the staff:

- 1) a **clef** (from the Latin *claves*, meaning “key”)
- 2) a **key signature**, which serves to indicate what note is the **tonic**

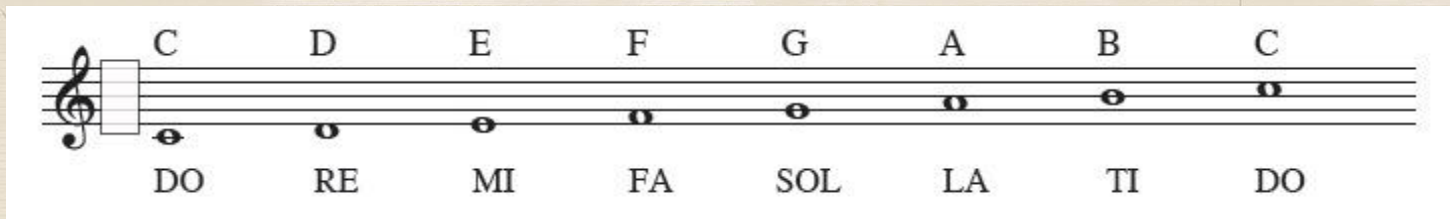
The two clefs in use today are the **treble** or “G-clef”



and the **bass** or “F-clef”

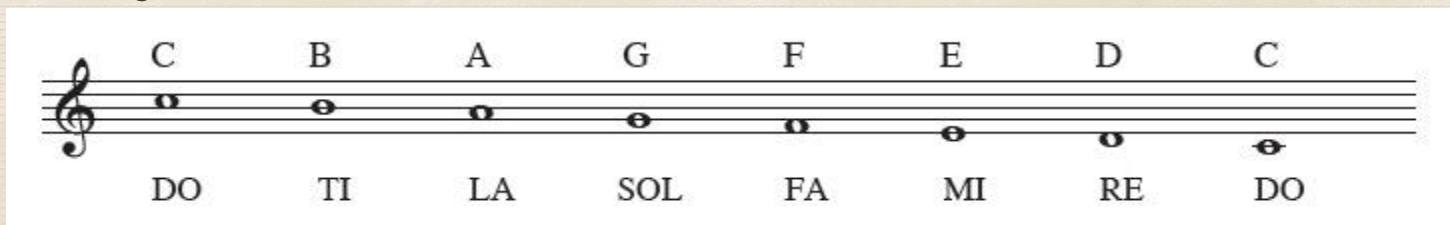


The key signature that contains **no** symbols of alteration, called **sharps** (#) or **flats** (b) indicates that the music that follows is likely to be in the **key of C Major** (or its **relative minor**, which we will discuss in a later unit).



These two pieces of information—the **clef**, and the **key signature**—allow us to assign letter names to the notes and to decide which note is the tonic=DO.

Using the ascending representation of the C-major scale above and the descending representation below in place of the staircase images, we can now practice the intervals of the major scale using real staff notation.



Sopranos and **Altos** commonly sing music notated in treble clef.

Basses and **Tenors** typically sing music notated in bass clef, although knowledge of the treble clef is also often required, especially of tenors. The C Major scale, ascending and descending, is shown below in bass clef.

The image displays two musical staves in bass clef, illustrating the C Major scale. The first staff shows the ascending scale, and the second staff shows the descending scale. Each note is represented by a whole note, and the corresponding letter name and solfège syllable are provided below the staff.

Scale	Letter Name	Solfège Syllable
Ascending	C	DO
	D	RE
	E	MI
	F	FA
	G	SOL
	A	LA
	B	TI
	C	DO
Descending	C	DO
	B	TI
	A	LA
	G	SOL
	F	FA
	E	MI
	D	RE
	C	DO

You are now ready to explore and master the
intervals of the
C MAJOR SCALE IN STAFF NOTATION

Please proceed to
PITCH EXERCISES 4.1 & 4.2

